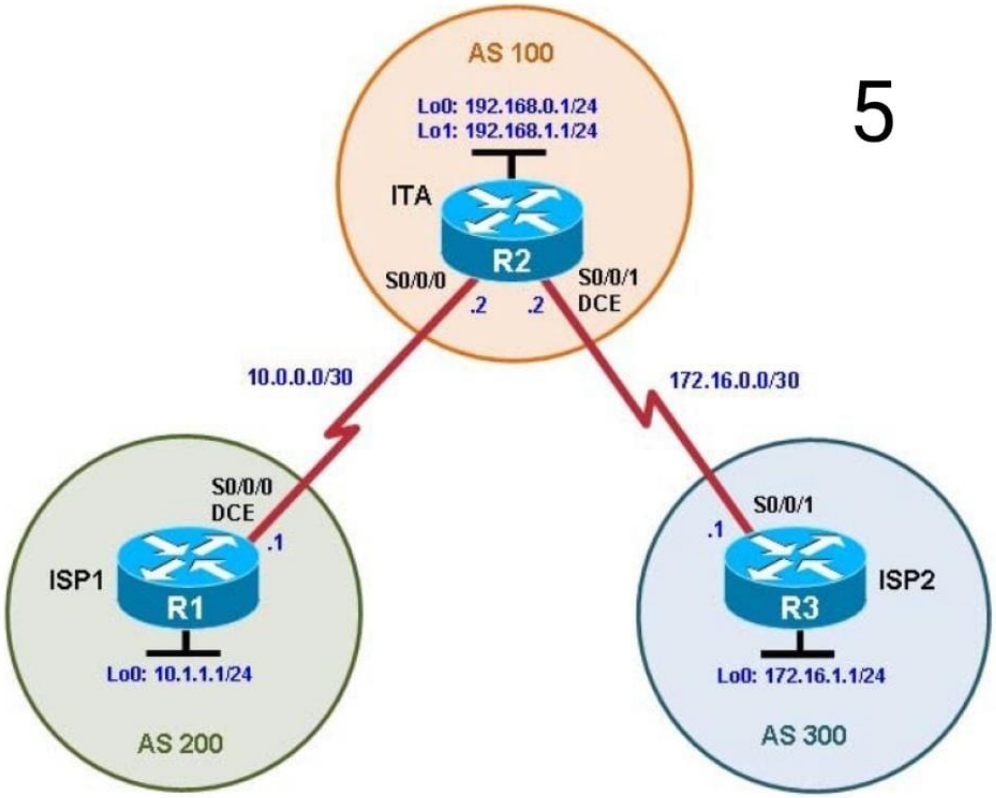

Step 1: Configure the Hostname and Interface Addresses

● Router R1 (ISP1) Configuration

```
Router R1 (hostname ISP1)

hostname ISP1

interface Lo0
 description ISP1 Internet Network
 ip address 10.1.1.1 255.255.255.0

interface Serial0/0/0
 description ISP1 -> ITA
 ip address 10.0.0.1 255.255.255.252
 clock rate 128000
 no shutdown

end
```

● Router R2 (ITA) Configuration

```
Router R2 (hostname ITA)

hostname ITA

interface Lo0
 description Core router network link 1
 ip address 192.168.0.1 255.255.255.0

interface Lo1
 description Core router network link 2
 ip address 192.168.1.1 255.255.255.0

interface Serial0/0/0
 description ITA -> ISP1
 ip address 10.0.0.2 255.255.255.252
 no shutdown

interface Serial0/0/1
 description ITA -> ISP2
 ip address 172.16.0.2 255.255.255.252
 clock rate 128000
 no shutdown

end
```

● Router R3 (ISP2) Configuration

```
Router R3 (hostname ISP2)

hostname ISP2

interface Lo0
 description ISP2 Internet Network
 ip address 172.16.1.1 255.255.255.0

interface Serial0/0/1
 description ISP2 -> ITA
 ip address 172.16.0.1 255.255.255.252
 no shutdown

end
```

Step 2: Configure BGP on the ISP Routers

● ISP1

```
ISP1(config)# router bgp 200
ISP1(config-router)# neighbor 10.0.0.2 remote-as 100
ISP1(config-router)# network 10.1.1.0 mask 255.255.255.0
```

● ISP2

```
ISP2(config)# router bgp 300
ISP2(config-router)# neighbor 172.16.0.2 remote-as 100
ISP2(config-router)# network 172.16.1.0 mask 255.255.255.0
```

Step 3: Configure BGP on the ITA Boundary Router

```
ITA(config)# router bgp 100
ITA(config-router)# neighbor 10.0.0.1 remote-as 200
ITA(config-router)# neighbor 172.16.0.1 remote-as 300
ITA(config-router)# network 192.168.0.0
ITA(config-router)# network 192.168.1.0

ITA# show ip route
```

Step 4: Verify BGP on the Routers

```
ITA# show ip bgp
ISP1# show ip bgp
ITA# show ip bgp neighbors
```

Step 5: Configure Route Filters

```
ISP2# show ip rout

ITA(config)# access-list 1 permit 192.168.0.0 0.0.1.255
ITA# clear ip bgp *

ISP2# show ip route
ISP1# show ip route
```

Step 6: Configure Primary and Backup Routes Using Floating Static Routes

```
ITA# show ip route

ITA(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.1 210
ITA(config)# ip route 0.0.0.0 0.0.0.0 172.16.0.1 220

ITA# show ip route
```

Configure Loopback on ISP1

```
ISP1# config t
ISP1(config)# interface loopback 100
ISP1(config-if)# ip address 192.168.100.1 255.255.255.0
```

Connectivity Test

```
ITA# show ip route
ITA# ping
ITA# ping 192.168.100.1 source 192.168.1.1
```

Step 7: Configure Primary and Backup Routes Using Default Network and Static Route

Remove Previous Static Routes

```
ITA(config)# no ip route 0.0.0.0 0.0.0.0 10.0.0.1 210
ITA(config)# no ip route 0.0.0.0 0.0.0.0 172.16.0.1 220
```

Advertise Default Network via BGP

```
ISP1(config)# router bgp 200
ISP1(config-router)# network 192.168.100.0
ISP1(config-router)# end
```

Configure Default Network

```
ITA# show ip route

ITA(config)# ip default-network 192.168.100.0
ITA# show ip route
```

Configure Backup Route

```
ITA(config)# ip route 0.0.0.0 0.0.0.0 172.16.0.1 220
ITA# show ip route
```

Refresh BGP and Verify

```
ITA# clear ip bgp 10.0.0.1
ITA# show ip route
```

Chatgpt format

Step 1: Configure hostname and interface addresses

Router R1 (ISP1)

```
enable
configure terminal

hostname ISP1

interface Loopback0
description ISP1 Internet Network
ip address 10.1.1.1 255.255.255.0
exit

interface Serial0/0/0
description ISP1 -> ITA
ip address 10.0.0.1 255.255.255.252
clock rate 128000
no shutdown
exit

end
```

Router R2 (ITA)

```
enable
configure terminal

hostname ITA

interface Loopback0
description Core router network link 1
ip address 192.168.0.1 255.255.255.0
exit

interface Loopback1
description Core router network link 2
ip address 192.168.1.1 255.255.255.0
exit

interface Serial0/0/0
description ITA -> ISP1
ip address 10.0.0.2 255.255.255.252
no shutdown
exit

interface Serial0/0/1
description ITA -> ISP2
ip address 172.16.0.2 255.255.255.252
clock rate 128000
no shutdown
exit

end
```

Router R3 (ISP2)

```
enable
configure terminal

hostname ISP2

interface Loopback0
description ISP2 Internet Network
ip address 172.16.1.1 255.255.255.0
exit

interface Serial0/0/1
description ISP2 -> ITA
ip address 172.16.0.1 255.255.255.252
no shutdown
exit

end
```

Step 2: Configure BGP on ISP Routers

ISP1

```
configure terminal
router bgp 200
neighbor 10.0.0.2 remote-as 100
network 10.1.1.0 mask 255.255.255.0
end
```

ISP2

```
configure terminal
router bgp 300
neighbor 172.16.0.2 remote-as 100
network 172.16.1.0 mask 255.255.255.0
end
```

Step 3: Configure BGP on ITA Router

```
configure terminal
router bgp 100
neighbor 10.0.0.1 remote-as 200
neighbor 172.16.0.1 remote-as 300
network 192.168.0.0
network 192.168.1.0
end
```

Verification

```
show ip route
```

Step 4: Verify BGP

```
show ip bgp
show ip bgp neighbors
```

Run on all routers if needed.

Step 5: Configure Route Filters

```
show ip route
configure terminal
access-list 1 permit 192.168.0.0 0.0.1.255
end
clear ip bgp *
```

Verification

```
show ip route
```

Step 6: Configure Floating Static Routes

```
show ip route
configure terminal
ip route 0.0.0.0 0.0.0.0 10.0.0.1 210
ip route 0.0.0.0 0.0.0.0 172.16.0.1 220
end
show ip route
```

Add loopback on ISP1

```
configure terminal
interface Loopback100
ip address 192.168.100.1 255.255.255.0
end
```

Test connectivity

```
show ip route
ping 192.168.100.1 source 192.168.1.1
```

Step 7: Default Network and Static Route

Remove floating routes

```
configure terminal
no ip route 0.0.0.0 0.0.0.0 10.0.0.1 210
no ip route 0.0.0.0 0.0.0.0 172.16.0.1 220
end
```

Configure default network on ISP1

```
configure terminal
router bgp 200
network 192.168.100.0
end
```

Configure default network on ITA

```
show ip route
configure terminal
ip default-network 192.168.100.0
ip route 0.0.0.0 0.0.0.0 172.16.0.1 220
end
```

Final verification

```
show ip route
clear ip bgp 10.0.0.1
show ip route
```